






Advanced Time Series Analysis and Forecasting

Seminar 13: Foundation models and conformal prediction



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material (1/2)

- **ACI**: Adaptive Conformal Inference
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARX**: AutoRegressive model with eXogenous variables
- **BET**: Bucharest Exchange Trading (index)
- **BG**: Bulgaria (Eurostat country code)
- **BH**: Benjamini–Hochberg (procedure)
- **CEE**: Central and Eastern Europe
- **CQR**: Conformalized Quantile Regression
- **CRPS**: Continuous Ranked Probability Score
- **CZ**: Czechia (Eurostat country code)
- **DM**: Diebold–Mariano (test)
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **EODHD**: EOD Historical Data (market-data provider)
- **EU**: European Union
- **FDR**: False Discovery Rate

Acronyms used in this material (2/2)

- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GW**: Giacomini–White (test of conditional predictive ability)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HICP**: Harmonised Index of Consumer Prices
- **HU**: Hungary (Eurostat country code)
- **LLM**: Large Language Model
- **MAE**: Mean Absolute Error
- **PID**: Proportional–Integral–Derivative (controller)
- **PL**: Poland (Eurostat country code)
- **RV**: Realised Variance
- **SK**: Slovakia (Eurostat country code)
- **VAT**: Value Added Tax
- **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's question and seminar structure

- Question of the seminar
  - **how good are zero-shot forecasts of pretrained models on our data, and how do we turn any forecast into intervals with controlled coverage when the data are dependent?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the Exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	split conformal by hand; the Beta law and the calibration size	Solved, Proposed	A1
A3, A4	ACI and quantile tracking by hand; the long-run bounds	Solved, Proposed	A3
A5, A6	CQR by hand; weighted conformal quantiles	Solved, Proposed	A5
A7, A8	the Chronos tokeniser; range limits and arcsinh scaling	Solved, Proposed	A7
B1, B2	Romanian load after the releases; calendar covariates in context	Solved, Proposed	B1
B3, B4	static, rolling and adaptive conformal for BET; the ACI design on the DAX	Solved, Proposed	B3
B5, B6	online CQR for Bitcoin RV forecasts; EnbPI on the load peak	Solved, Proposed	B5
B7, B8	CEE inflation with Holm and BH; VaR of the WIG20	Solved, Proposed	B7
C1, C2	a pre-registered inflation forecast; what is wrong in an AI answer?	Proposed	B5, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in class
- Part A: calculations and derivations on paper; Part B: real data, solved in the notebook, each ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data used

Series	Source	Frequency, period
Romanian electricity load	Energy-Charts (ENTSO-E)	hourly, 2022–2026
HICP annual inflation, EU countries	Eurostat	monthly, 2000–2026
Bitcoin realised variance	Binance, 5-minute returns	daily, 2018–2026
BET, DAX, WIG20	EODHD	daily, 2000–2026

- Returns are 100 times log differences
- In the notebook; no account or key is needed
  - `fm_forecast`, `conformal_quantile`, `online_threshold`, `cqr`, `enbpi`
  - `dm_test`, `holm`, `bh`, `kupiec`, `christoffersen`

Prerequisites for Today (1/23): foundation models

- A **foundation model** is a network $f_{\hat{\theta}}$ pretrained once on a corpus of many series; $\hat{\theta}$: its estimated weights
- **Zero-shot** use: the forecast of a new series $y_1, \ldots, y_T$ is the network applied to its last C values; nothing is estimated on the new series

$$\hat{y}_{T+1:T+H} = f_{\hat{\theta}}(y_{T-C+1:T})$$

  - C: the context length; H: the horizon; T: the forecast origin
- **Outputs**: quantile forecasts $\hat{q}_\tau$ at fixed levels τ
  - $\hat{q}_\tau$: the value below which the outcome falls with probability τ, according to the model
  - Chronos-Bolt, TimesFM 2.5, TiRex: 10%–90% (deciles); Chronos-2: 1%–99% Ansari et al. (2025)
- **Covariates in context** (Chronos-2): past values of the covariates, and future values of those known in advance (calendar), are passed with the target
  - their effect is inferred inside the forward pass, not estimated
- **Contamination**: the test window may be in the training data
  - evaluate after the release of every model (here: from 1 November 2025)

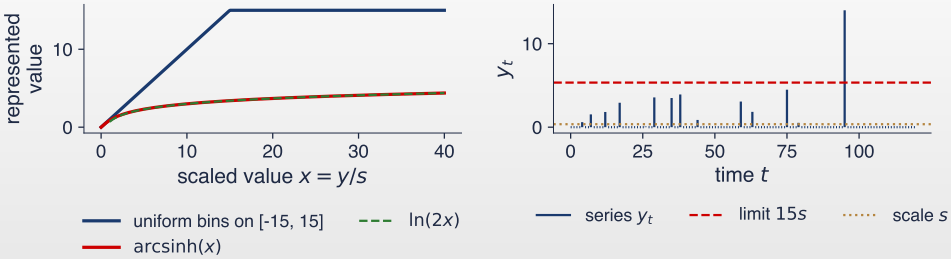
Prerequisites for Today (2/23): the Chronos tokeniser

- **Mean scaling** Ansari et al. (2024): each context value is divided by the mean absolute value of the context

$$s = \frac{1}{C}\sum_{t=1}^{C}|y_t|, \qquad \tilde{y}_t = y_t/s$$

  - s: the scale; $\tilde{y}_t$: the scaled value
- **Quantisation**: 4093 bin centres equally spaced on $[-15, 15]$; each scaled value is replaced by the nearest centre, and the index of that centre is its token
  - distance between neighbouring centres: $30/4092$ in scaled units, $30s/4092$ in data units
  - **dequantised value**: the centre times s; the quantisation error is at most half the spacing
- **Range limit**: values beyond $15s$ are truncated to the last centre and cannot be forecast
  - Chronos-2 uses $\operatorname{arcsinh}(y/s)$ instead, with $\operatorname{arcsinh}(x) = \ln\left(x + \sqrt{x^2+1}\right)$: close to x near 0, close to $\ln(2x)$ for large x

Prerequisites for Today (3/23): the range of the tokeniser



- Left: uniform bins represent a scaled value $x = y/s$ only up to 15; $\operatorname{arcsinh}(x)$ has no ceiling and grows like $\ln(2x)$
  - $\operatorname{arcsinh}$ compresses large values instead of cutting them: a value 1000 times the scale becomes $\operatorname{arcsinh}(1000) \approx 7.6$
- Right: a simulated intermittent series (zero on most days); scale $s = 0.36$, limit $15s = 5.34$
  - the rare value 14 is 39 times the scale: with mean scaling it lies outside the range and cannot be forecast
  - many zeros make s small, so the limit $15s$ is low

Prerequisites for Today (4/23): scores for quantile forecasts

- **Pinball loss** of a quantile forecast $\hat{q}_\tau$ for the outcome y

$$\rho_\tau(u) = u\big(\tau - 1\{u < 0\}\big), \qquad u = y - \hat{q}_\tau$$

  - $1\{\cdot\}$: 1 if the condition holds, 0 otherwise; an outcome above the quantile costs τu, one below costs $(1-\tau)|u|$
  - its expected value is minimised by the true τ-quantile

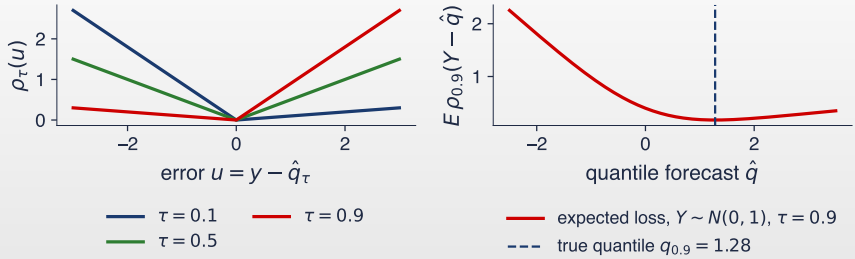
- **CRPS** approximated from K quantile levels $\tau_1, \ldots, \tau_K$ (here the nine deciles)

$$\mathrm{CRPS} \approx \frac{2}{K} \sum_{k=1}^{K} \rho_{\tau_k}\left(y - \hat{q}_{\tau_k}\right)$$

  - it scores the whole predictive distribution, in the units of y; lower is better

- **MAE** of the median: the mean of $|y - \hat{q}_{0.5}|$

Prerequisites for Today (5/23): the pinball loss and interval scores



- Left: $\rho_\tau(u)$ has slope τ for $u > 0$ and $-(1-\tau)$ for $u < 0$; for $\tau = 0.5$ it is half the absolute error
- Right: for $Y \sim N(0,1)$ and $\tau = 0.9$ the expected loss is smallest at the true quantile $q_{0.9} = 1.28$
- **Coverage** and **mean width** of intervals $[l_t, u_t]$ over n forecasts

$$\widehat{\text{cov}} = \frac{1}{n}\sum_{t=1}^{n} 1\{l_t \le y_t \le u_t\}, \qquad \bar{w} = \frac{1}{n}\sum_{t=1}^{n}(u_t - l_t)$$

  - a good interval has $\widehat{\text{cov}}$ close to the nominal level (e.g. 0.80 for the 10–90% band) with the smallest $\bar{w}$

Prerequisites for Today (6/23): the benchmark forecasts

- **Weekly naive** for hourly load: $\hat{y}_{d,h} = y_{d-7,h}$
  - $y_{d,h}$: the load of day d at hour h; the forecast repeats the same hour one week earlier
- **Expert ARX** Ziel and Weron (2018): one least-squares regression per hour h, on a rolling window

$$y_{d,h} = \beta_0 + \beta_1 y_{d-1,h} + \beta_2 y_{d-2,h} + \beta_3 y_{d-7,h} + \beta_4 \underline{y}_{d-1} + \beta_5 \overline{y}_{d-1} + \beta_6 y_{d-1,24} + \sum_j \delta_j D_{j,d} + \varepsilon_{d,h}$$

  - $\underline{y}_{d-1}$, $\overline{y}_{d-1}$: minimum and maximum load of the previous day; $y_{d-1,24}$: its last hour
  - $D_{j,d}$: 1 if day d is a Monday, a Saturday or a Sunday (j), 0 otherwise; lags 1, 2 and 7: persistence and the weekly cycle
  - quantiles: the point forecast plus the empirical quantiles of the recent errors at the same hour
- **AR(p)** for inflation, with p chosen by the AIC

$$y_t = c + \sum_{j=1}^{p} \phi_j y_{t-j} + \varepsilon_t, \qquad \mathrm{AIC}(p) = T \ln \hat{\sigma}_p^2 + 2(p+1)$$

  - ϕ_j: the autoregressive coefficients; $\hat{\sigma}_p^2$: the residual variance of the AR(p); T: the sample size
  - the penalty $2(p+1)$ counts the coefficients; the order with the smallest AIC is kept

Prerequisites for Today (7/23): comparing two models

- **DM–HLN test** (Chapter 1): are the expected losses of two models A and B equal? The statistic is on the next slide
  - $t < 0$: model A has the lower loss
- **Across series**: the ratio $r_s = L_s(\text{model})/L_s(\text{baseline})$ on series s, combined by the geometric mean Fleming and Wallace (1986)

$$\bar{r}_{\text{geo}} = \exp\left(\frac{1}{S}\sum_{s=1}^{S} \ln r_s\right)$$

  - L_s: a loss on series s; S: the number of series; $\bar{r}_{\text{geo}} < 1$: the model beats the baseline on average

Prerequisites for Today (8/23): the DM–HLN statistic

- Loss differential at origin t and its mean over n origins; H_0: $E[d_t] = 0$

$$d_t = L(e_{A,t}) - L(e_{B,t}), \qquad \bar{d} = \frac{1}{n}\sum_{t=1}^{n} d_t$$

  - $e_{A,t}$, $e_{B,t}$: the forecast errors of models A and B; L: the loss (absolute error, pinball, CRPS)

- **HAC variance** of $\bar{d}$: the autocovariances of d_t up to lag ℓ are added

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{\ell} w_j \hat{\gamma}_j, \qquad \hat{\gamma}_j = \frac{1}{n}\sum_{t=j+1}^{n} (d_t - \bar{d})(d_{t-j} - \bar{d})$$

  - $\hat{\gamma}_j$: the autocovariance at lag j; here $w_j = 1$ and $\ell = h - 1$ (h: the horizon), or Bartlett weights $w_j = 1 - j/(\ell + 1)$; B7 uses $\ell = 5$

- **Statistic** Diebold and Mariano (1995) with the small-sample correction Harvey, Leybourne and Newbold (1997)

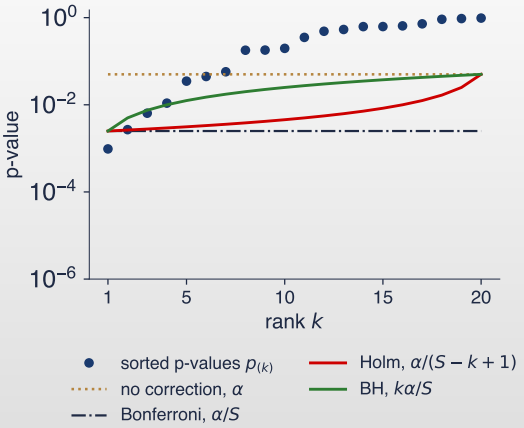
$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/n}}, \qquad \mathrm{DM}_{\mathrm{HLN}} = \mathrm{DM}\sqrt{\frac{n + 1 - 2h + h(h-1)/n}{n}}$$

  - reference distribution: Student-t with $n - 1$ degrees of freedom; p-value $= 2\Pr\{t_{n-1} > |\mathrm{DM}_{\mathrm{HLN}}|\}$

Prerequisites for Today (9/23): tests on many series

- With S tests at 5% each, about $0.05S$ false rejections are expected under H_0; $p_{(1)} \leq \cdots \leq p_{(S)}$: the sorted p-values
- **Holm** Holm (1979) controls the FWER, the probability of at least one false rejection
  - reject while $p_{(k)} \leq \alpha/(S-k+1)$, $k = 1, 2, \ldots$; stop at the first failure
- **Benjamini–Hochberg** Benjamini and Hochberg (1995) controls the FDR, the expected share of false rejections among the rejections
  - reject the k^* smallest p-values, $k^* = \max\{k : p_{(k)} \leq k\alpha/S\}$
- **Pooled test**: average d_t across series at each origin, then one DM test on this average

Prerequisites for Today (10/23): three corrections on 20 tests



- Simulated: 20 tests, 6 with a true effect; $\alpha = 0.05$
  - a p-value is rejected when it lies below its line
- Lines: the thresholds of the previous slide, against the rank k; Bonferroni: α/S for every k
  - Holm stops at the first p-value above its line
  - BH rejects all p-values up to the last one below its line
- Rejections here: 6 without correction, 1 Bonferroni, 1 Holm, 3 BH
  - FWER control is stricter than FDR control: Holm rejects fewer hypotheses than BH

Prerequisites for Today (11/23): split conformal

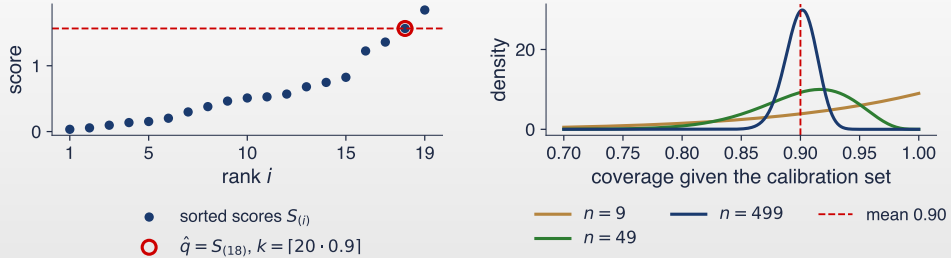
- **Exchangeable** data $Z_i = (X_i, Y_i)$, $i = 1, \ldots, n+1$: the joint law does not change when the points are reordered (i.i.d. is a special case)
  - X_i: predictors; Y_i: the value to be covered; point $n+1$ is the test point
- A **nonconformity score** $s(x, y)$ is large when y is unusual given x, e.g. $|y - \hat{f}(x)|$ with $\hat{f}$ a point forecast
- **Split conformal** Lei et al. (2018): with the n calibration scores $S_i = s(X_i, Y_i)$ and a miss rate α

$$k = \lceil (n+1)(1-\alpha) \rceil, \qquad \hat{q} = S_{(k)}, \qquad \hat{C}(x) = \{y : s(x, y) \leq \hat{q}\}$$

  - $S_{(k)}$: the k-th smallest score; $\lceil \cdot \rceil$: rounding up; $\hat{q} = +\infty$ if $k > n$; with $|y - \hat{f}(x)|$ the set is $\hat{f}(x) \pm \hat{q}$
- **Guarantee** under exchangeability (no ties)

$$1 - \alpha \leq \Pr\{Y_{n+1} \in \hat{C}(X_{n+1})\} \leq 1 - \alpha + \frac{1}{n+1}$$

Prerequisites for Today (12/23): the conformal threshold and its Beta law

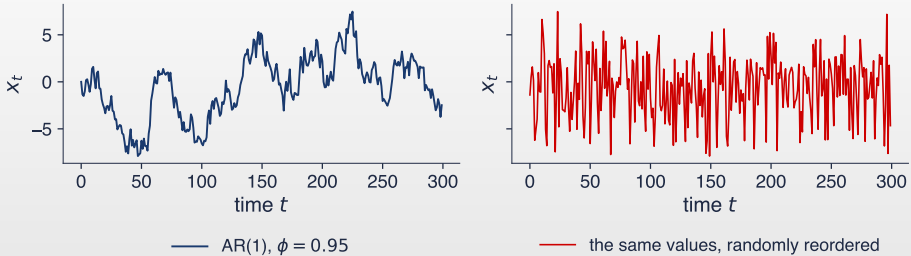


- Left: 19 simulated absolute residuals; with $\alpha = 0.1$, $\hat{q}$ is the 18th smallest score (here 1.57)
- Right: the realised coverage C given the calibration set has the Beta$(k, n+1-k)$ density

$$f(c) = \frac{c^{k-\mathbf{1}}(1-c)^{n-k}}{B(k, n+1-k)}, \qquad \mathbb{E}C = \frac{k}{n+1}, \qquad \mathrm{Var}(C) = \frac{k(n+1-k)}{(n+1)^{\mathbf{2}}(n+2)}$$

  - $B(\cdot,\cdot)$: the Beta function, which makes the density integrate to one; standard deviation for $n = 9$, 49, 499: 0.090, 0.042, 0.013

Prerequisites for Today (13/23): exchangeability and time series



- Left: a simulated AR(1), $x_t = 0.95x_{t-\mathbf{1}} + \varepsilon_t$; right: the same 300 values in a random order
  - lag-1 autocorrelation: 0.96 before, 0.01 after the reordering
  - the two paths have visibly different joint laws: an autocorrelated series is not exchangeable
- **Exchangeable**: $(Z_{\pi(\mathbf{1})}, \ldots, Z_{\pi(n+\mathbf{1})})$ has the same law as $(Z_{\mathbf{1}}, \ldots, Z_{n+\mathbf{1}})$ for every permutation π of the indices
  - for dependent data the split conformal guarantee holds only approximately, if the dependence fades with the lag (mixing)

Prerequisites for Today (14/23): CQR and weighted conformal

- **CQR** Romano, Patterson and Candès (2019): start from quantile forecasts $\hat{q}_{\mathrm{lo}}(x)$, $\hat{q}_{\mathrm{hi}}(x)$ (e.g. 10% and 90%)

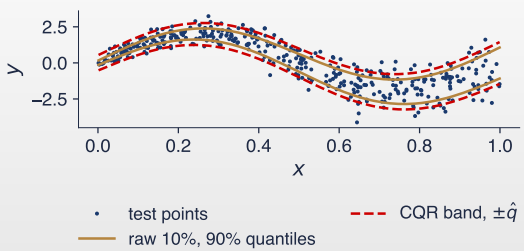
$$s(x,y) = \max\{\hat{q}_{\mathrm{lo}}(x) - y,\ y - \hat{q}_{\mathrm{hi}}(x)\}, \qquad \hat{C}(x) = [\hat{q}_{\mathrm{lo}}(x) - \hat{q},\ \hat{q}_{\mathrm{hi}}(x) + \hat{q}]$$

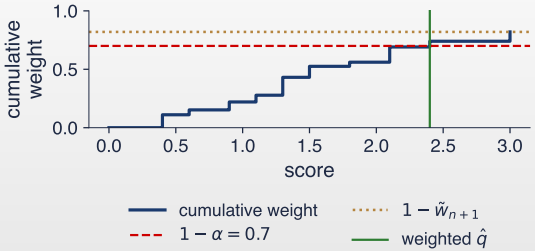
  - the score is the signed distance outside the band (negative inside); $\hat{q}$: the split conformal threshold of these scores
  - $\hat{q} > 0$ widens the band, $\hat{q} < 0$ narrows it
- **Weighted conformal** Barber et al. (2023): weights $w_i \in [0,1]$ fixed in advance (e.g. larger for recent points), $w_{n+1} = 1$ for the test point

$$\tilde{w}_i = \frac{w_i}{1 + \sum_{j=1}^{n} w_j}, \qquad \hat{q} = Q_{1-\alpha}\Big(\sum_{i=1}^{n} \tilde{w}_i\, \delta_{S_i} + \tilde{w}_{n+1}\, \delta_{+\infty}\Big)$$

  - δ_a: a point mass at a; $Q_{1-\alpha}$: the $(1-\alpha)$ quantile; the test point is placed at $+\infty$ with mass $\tilde{w}_{n+1}$
  - in practice: sort the scores, add their weights $\tilde{w}$ in that order, stop when the sum reaches $1-\alpha$

Prerequisites for Today (15/23): CQR and weighted conformal: a numerical example





- Left: simulated data whose noise grows with x; the raw 10%–90% quantile forecasts are too narrow (coverage 0.57 on test points)
  - CQR with 400 calibration points: $\hat{q} = 0.38$, coverage 0.85; the band keeps its varying width
- Right: ten scores with weights $w_i = 0.85^{\mathbf{11}-i}$; cumulative normalised weight of the sorted scores; $\alpha = 0.3$
  - the curve stops at $1 - \tilde{w}_{n+\mathbf{1}} = 0.820$, the rest being the mass at $+\infty$; it reaches 0.7 at $\hat{q} = 2.4$
  - **effective sample size** $n_{\text{eff}} = (\sum_i w_i)^{\mathbf{2}} / \sum_i w_i^{\mathbf{2}} = 8.3$: the number of equally weighted points with the same information

Prerequisites for Today (16/23): conformal methods online

- **Online frame**: scores S_t from out-of-sample forecasts; at time t the threshold q_t uses only $S_1, \ldots, S_{t-1}$
  - miss indicator $\mathrm{err}_t = 1\{S_t > q_t\}$: 1 if Y_t falls outside the interval
  - static: calibrated once; rolling: the conformal quantile of the last w scores
- **ACI** Gibbs and Candès (2021): q_t is the conformal quantile at a working level α_t, updated after each observation

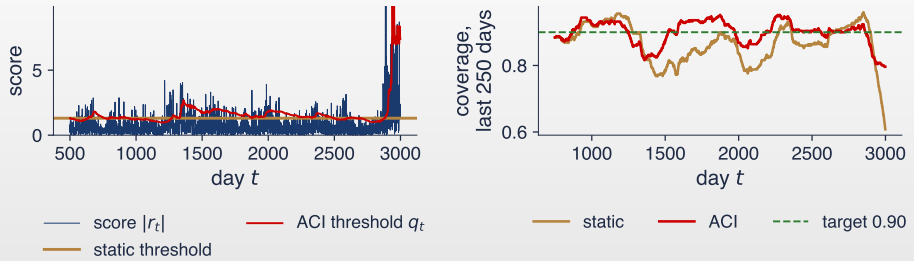
$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t), \qquad \left|\frac{1}{T}\sum_{t=1}^{T} \mathrm{err}_t - \alpha\right| \leq \frac{\max\{\alpha_1, 1-\alpha_1\} + \gamma}{\gamma T}$$

  - $\gamma > 0$: the step; α_1: the starting level; a miss lowers α_t (wider interval); $\alpha_t \leq 0$ gives $q_t = +\infty$
- **Quantile tracking** Angelopoulos, Candès and Tibshirani (2023): the threshold itself moves; with scores in $[0, B]$

$$q_{t+1} = q_t + \eta(\mathrm{err}_t - \alpha), \qquad \left|\frac{1}{T}\sum_{t=1}^{T}(\mathrm{err}_t - \alpha)\right| \leq \frac{B + \eta}{\eta T}$$

  - $\eta > 0$: the step, in score units; conformal PID adds an integrator of past errors and a model that forecasts the next score

Prerequisites for Today (17/23): ACI under volatility clustering



- Simulated GARCH(1,1) returns $r_t = \sigma_t z_t$, $\sigma_t^2 = 0.02 + 0.08 r_{t-1}^2 + 0.90 \sigma_{t-1}^2$, z_t Student-t; score $|r_t|$, $\alpha = 0.1$
  - static threshold: calibrated once on the first 500 days; ACI: $\gamma = 0.005$ on the last 500 scores
- **Local coverage** over the last $W = 250$ days: $\frac{1}{W} \sum_{s=t-W+1}^{t} (1 - \mathrm{err}_s)$
  - static: overall 0.847, local from 0.608 to 0.960; ACI: overall 0.893, local from 0.796 to 0.952 (2500 days)
  - in turbulent periods the static threshold is too low; ACI raises q_t after misses and recovers the frequency

Prerequisites for Today (18/23): EnbPI

- **EnbPI** Xu and Xie (2021): B models fitted on block-bootstrap samples of the training period
  - block bootstrap: B series of blocks of b consecutive training points, drawn with replacement
  - S_b: the set of points in sample b; blocks keep the short-run dependence (B6: blocks of 7 days)
- **Ridge regression** on each sample: least squares with a penalty on the size of the coefficients

$$\hat{\beta}^{(b)} = \arg\min_{\beta} \sum_{i \in S_b} (y_i - x_i'\beta)^2 + \lambda \|\beta\|^2$$

  - $\lambda > 0$: the penalty; $\|\beta\|^2$: the sum of squared coefficients; $\hat{f}^{(b)}(x) = x'\hat{\beta}^{(b)}$
- **Leave-one-out residuals** (for point i, only the models whose sample excluded i) and the **interval**

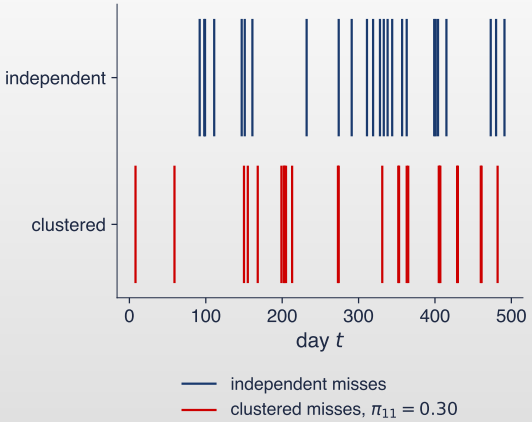
$$\hat{\varepsilon}_i = \left| y_i - \frac{1}{|\{b : i \notin S_b\}|} \sum_{b:\, i \notin S_b} \hat{f}^{(b)}(x_i) \right|, \qquad \hat{C}_t = \hat{f}(x_t) \pm Q_{1-\alpha}\left(\hat{\varepsilon}_{t-w}, \ldots, \hat{\varepsilon}_{t-1}\right)$$

  - $\hat{f}$: the mean of the B models; $Q_{1-\alpha}$: the $(1-\alpha)$ empirical quantile of the last w absolute residuals
  - after each new observation its residual enters the window and the oldest one leaves

Prerequisites for Today (19/23): coverage tests

- **Kupiec** Kupiec (1995): tests H_0: $\Pr\{\mathrm{err}_t = 1\} = \alpha$ on the miss sequence
  - a small p-value: the miss rate differs from α (statistic on the next slide)
- **Christoffersen** Christoffersen (1998): adds independence, against a first-order Markov alternative (a miss today changes the chance of a miss tomorrow)
  - a small p-value: wrong frequency or clustered misses
- **Local coverage**: the share of hits over a moving window (Gibbs and Candès use 500 days)
  - or coverage by classes of a variable known at the origin

Prerequisites for Today (20/23): the Kupiec and Christoffersen statistics



- **Kupiec**: x misses in T periods, $\hat{\pi} = x/T$

$$LR_{uc} = -2 \ln \frac{(1-\alpha)^{T-x}\alpha^{x}}{(1-\hat{\pi})^{T-x}\hat{\pi}^{x}} \sim \chi^2(1)$$

- **Christoffersen**: n_{ij} = number of days with state i followed by state j (1 = miss)

$$\hat{\pi}_{01} = \frac{n_{01}}{n_{00} + n_{01}}, \qquad \hat{\pi}_{11} = \frac{n_{11}}{n_{10} + n_{11}}$$

  - LR_{ind}: likelihood ratio of one miss probability against $\hat{\pi}_{01} \neq \hat{\pi}_{11}$
  - $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$: frequency and independence tested together

- Simulated, $T = 500$, $\alpha = 0.05$
  - independent: 25 misses, Kupiec $p = 1.00$, LR_{cc} $p = 0.970$
  - clustered: 25 misses, $\hat{\pi}_{11} = 0.280$, Kupiec $p = 1.00$, LR_{cc} $p < 0.001$

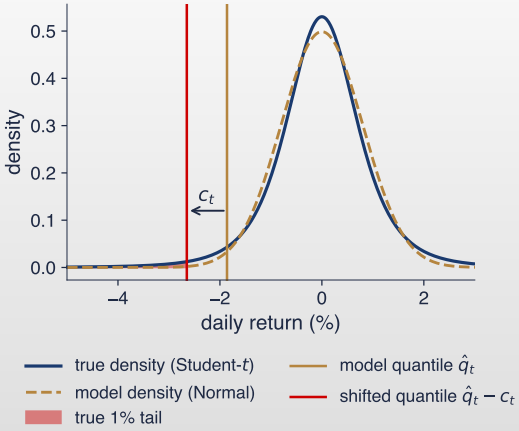
Prerequisites for Today (21/23): VaR from a quantile forecast

- **VaR** (Chapter 9): $\mathrm{VaR}_\alpha = -q_\alpha$, where q_α is the α-quantile of the return distribution
  - VaR 1%: $\alpha = 0.01$; the target exceedance rate is α; the quantile loss is the pinball loss ρ_α
- **One-sided conformal shift** of a quantile forecast $\hat{q}_t$

$$S_t = \hat{q}_t - y_t, \qquad \mathrm{VaR}_t = -(\hat{q}_t - c_t)$$

  - c_t: the online threshold of the scores S_t at level α, so that $\Pr\{S_t > c_t\} \approx \alpha$; $c_t > 0$ moves the quantile down (a larger VaR)

Prerequisites for Today (22/23): a one-sided conformal shift of the VaR



- Simulated: true returns Student-t with 4 degrees of freedom and unit variance; the model is $N(0, 0.8^2)$
  - model 1% quantile $\hat{q}_t = -1.86$: it is exceeded with probability 2.9%, not 1%
- Scores S_t of the previous slide: positive when the return falls below the forecast quantile
  - here $c_t = 0.79$ moves the quantile to the true -2.65: VaR 1% rises from 1.86 to 2.65
- The shift corrects the exceedance rate; it adds no information about when the tail risk is high

Prerequisites for Today (23/23): the volatility benchmarks

- **GARCH(1,1)**-t Bollerslev (1986) (Chapter 9) for returns r_t (in %), with zero mean

$$r_t = \sigma_t z_t, \qquad \sigma_t^2 = \omega + a_1 r_{t-1}^2 + b_1 \sigma_{t-1}^2, \qquad z_t \sim t_\nu \text{ (unit variance)}$$

  - $\omega > 0$, $a_1, b_1 \geq 0$, $a_1 + b_1 < 1$: persistence; ν: the degrees of freedom (tail thickness); estimated by maximum likelihood
  - VaR: $\mathrm{VaR}_{\alpha,t} = -\sigma_t q_\alpha(z)$, with $q_\alpha(z)$ the α-quantile of z_t

- **Realised variance** of day t from M intraday returns $r_{t,i}$ (5-minute returns: $M = 288$ for Bitcoin)

$$RV_t = \sum_{i=1}^{M} r_{t,i}^2, \qquad x_t = \ln RV_t$$

- **HAR** Corsi (2009): the next-day log variance on its daily, weekly and monthly averages

$$x_{t+1} = c + \beta_d x_t + \beta_w \frac{1}{5} \sum_{j=0}^{4} x_{t-j} + \beta_m \frac{1}{22} \sum_{j=0}^{21} x_{t-j} + u_{t+1}$$

  - three horizons of traders (day, week, month) give long-lasting persistence with only four coefficients

A1: split conformal by hand [Solved]

Task

- 19 absolute calibration residuals, sorted: 0.2; 0.3; 0.4; 0.5; 0.6; 0.7; 0.8; 0.9; 1.0; 1.1; 1.2; 1.3; 1.5; 1.6; 1.8; 2.0; 2.1; 2.6; 3.2; a new point forecast $\hat{f} = 10$; $\alpha = 0.1$.
- 1. Compute k and $\hat{q}$.
- 2. Write the 90% interval.
- 3. Give the lower and upper bounds of the marginal coverage under exchangeability.
- Report: two numbers; one interval; two numbers and one sentence.

Solution

- 1. $k = \lceil 20 \cdot 0.9 \rceil = 18$; $\hat{q} = S_{(\mathbf{18})} = 2.6$
- 2. $[10 - 2.6, 10 + 2.6] = [7.4, 12.6]$
- 3. $0.90 \leq \text{coverage} \leq 0.90 + 1/20 = 0.95$
- Interpretation: with 19 points the interval uses the second-largest residual; the guarantee is over new calibration sets as well as new points

Back Exercise list

A2: the size of the calibration set [Proposed]

Solution discussed in class

Task

- $\alpha = 0.1$; the law of the coverage given the calibration set: Prerequisites (12/23). Model: A1.
- 1. For $n = 19$, 99 and 999 give k, the mean and the standard deviation of the coverage.
- 2. For each n, compute the probability that the realised coverage is below 0.85.
- 3. Find the smallest n whose standard deviation is at most 0.01.
- Report: three triples; three probabilities; one number.

A3: ACI by hand [Solved]

Task

- $\alpha = 0.1$, $\gamma = 0.01$, $\alpha_{\mathbf{1}} = 0.1$; observed misses $\mathrm{err}_{\mathbf{1}}, \ldots, \mathrm{err}_{\mathbf{5}} = 1, 1, 0, 1, 0$.
- 1. Compute $\alpha_{\mathbf{2}}, \ldots, \alpha_{\mathbf{6}}$.
- 2. Compute the bound on $|\frac{\mathbf{1}}{T}\sum \mathrm{err}_t - \alpha|$ for $T = 1000$.
- 3. After how many consecutive misses from $\alpha_{\mathbf{1}}$ is the interval infinite? After how many consecutive hits is it empty?
- Report: five numbers; one number; two integers and one sentence.

Solution

- 1. A miss subtracts $\gamma(1-\alpha) = 0.009$, a hit adds $\gamma\alpha = 0.001$: 0.091; 0.082; 0.083; 0.074; 0.075
- 2. $(\max\{0.1, 0.9\} + 0.01)/(0.01 \cdot 1000) = 0.091$
- 3. $\lceil 0.1/0.009 \rceil = 12$ misses; $\lceil 0.9/0.001 \rceil = 901$ hits
- Interpretation: ACI reacts nine times faster to a miss than to a hit; about a dozen consecutive misses lead to an interval equal to the whole real line, which keeps the long-run frequency

Back Exercise list

A4: quantile tracking by hand [Proposed]

Solution discussed in class

Task

- Scores in $[0, 5]$; $\alpha = 0.1$, $\eta = 0.05$, $q_1 = 1$; misses $1, 0, 0, 1, 1$. Model: A3.
- 1. Compute $q_2, \ldots, q_6$.
- 2. Compute the bound $(B + \eta)/(\eta T)$ for $T = 2000$.
- 3. Prove that q_t stays in $[-\eta, B + \eta]$ if it starts inside $[0, B]$.
- Report: five numbers; one number; a three-line proof.

A5: CQR by hand [Solved]

Task

- 8 calibration points with quantile forecasts and outcomes $(\hat{q}_{\mathrm{lo}}, \hat{q}_{\mathrm{hi}}, y)$: (1, 3, 2), (2, 4, 4.6), (0.5, 2.5, 0.2), (1.5, 3, 2.5), (3, 5, 5.4), (2.5, 4, 3.1), (1, 2.5, 2.2), (2, 4.5, 1.6); $\alpha = 0.2$.
- 1. Compute the eight CQR scores.
- 2. Compute k and $\hat{q}$.
- 3. Give the conformal interval for a new point with $(\hat{q}_{\mathrm{lo}}, \hat{q}_{\mathrm{hi}}) = (2, 4)$ and the raw coverage on the calibration set.
- Report: eight numbers; two numbers; one interval and one share.

Solution

- 1. -1.0; 0.6; 0.3; -0.5; 0.4; -0.6; -0.3; 0.4
- 2. $k = \lceil 9 \cdot 0.8 \rceil = 8$, $\hat{q} = 0.6$ (the largest score)
- 3. $[2 - 0.6, 4 + 0.6] = [1.4, 4.6]$; raw coverage 0.50
- Interpretation: the raw 80% band covers only half of the points; CQR widens it by the largest miss; a negative $\hat{q}$ would narrow an over-wide band

Back Exercise list

A6: weighted conformal quantiles [Proposed]

Solution discussed in class

Task

- Six scores in time order: 2.0, 0.5, 1.5, 3.0, 1.0, 2.5; weights $w_i = 0.8^{7-i}$, $w_7 = 1$. Model: A5.
- 1. Compute the normalised weights $\tilde{w}_1, \ldots, \tilde{w}_7$.
- 2. Compute the weighted and the unweighted conformal thresholds for $\alpha = 0.3$.
- 3. Repeat for $\alpha = 0.2$ and explain the result with the mass $\tilde{w}_7$.
- Report: seven weights; two thresholds; two thresholds and one sentence.

A7: the Chronos tokeniser by hand [Solved]

Task

- Context of eight values: 100, 104, 98, 102, 106, 110, 96, 104; 4093 centres equally spaced on $[-15, 15]$.
- 1. Compute the scale s and the bin width in scaled units and in data units.
- 2. Find the token of the last value (centres indexed $0, \ldots, 4092$), its dequantised value and the error.
- 3. Give the largest value the model can forecast, and say what happens to a future value of 1700.
- Report: three numbers; three numbers; one number and one sentence.

Solution

- 1. $s = 102.5$; width $30/4092 = 0.00733$, i.e. 0.751 in data units
- 2. $104/s$ is closest to centre 2184; dequantised 103.70, error 0.30
- 3. $15s = 1537.5$; 1700 scales to $16.59 > 15$: clipped to the last bin, it cannot be forecast
- Interpretation: the resolution is relative to the context mean; a tenfold jump can be represented, a sixteenfold one cannot

Back Exercise list

A8: range limits and arcsinh scaling [Proposed]

Solution discussed in class

Task

- Context: seven values of 50 and one of 70; possible future values 90, 400, 1500. Model: A7.
- 1. Compute s and the scaled future values; say which ones Chronos cannot reach.
- 2. Compute $\operatorname{arcsinh}$ of the scaled values (the Chronos-2 input transform) and explain why a quantile head on this scale has no hard limit.
- 3. Name a Romanian series where this matters.
- Report: four numbers and one list; three numbers and one sentence; one example.

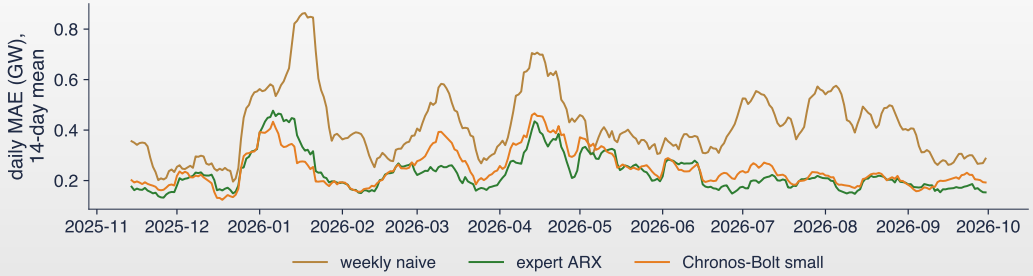
Back Exercise list

B1: Romanian load after the model releases [Solved]

- **Question**: is a zero-shot foundation model as accurate as the expert model for day-ahead Romanian load, on days it cannot have seen?
- **Inputs**: Energy-Charts hourly load, forecast days 1 November 2025 – 30 September 2026 (334 days); weekly naive; expert ARX of Ziel and Weron (Chapter 1); Chronos-Bolt small with 2048 hours of context
- **Tasks**
  1. Compute the MAE of the median and the CRPS over the deciles for the three models.
  2. Compute the coverage of the 10–90% band.
  3. Run the DM–HLN test of each model against the expert ARX on daily MAE.
  4. Interpretation: is the zero-shot model good enough to replace the expert model?
- **Report**: a table of three models and three scores, two DM statistics, one sentence
- Notebook: section B1

Back Exercise list

B1: solution [Solved]



- MAE (GW): weekly naive 0.412, expert ARX 0.227, Chronos-Bolt 0.246; CRPS 0.330, 0.184, 0.197; coverage of 80% bands 76.8%, 78.0%, 81.5%
- DM against the expert ARX: Chronos-Bolt $t = 3.06$ ($p = 0.002$); weekly naive $t = 12.69$
- Interpretation: not yet: on days it cannot have seen, Chronos-Bolt is 8% less accurate than the expert model, a significant gap, but it needs no estimation and its bands are better calibrated: a strong default, not a replacement

ATS_ch13_seminar

B2: calendar covariates in context [Proposed]

- **Question**: does Chronos-2 learn the effect of public holidays from the context alone?
- **Inputs**: the B1 window; Chronos-2 univariate and with two covariates passed for the context and for the forecast day: a holiday flag and a weekend flag; model: B1
- **Tasks**
  1. Compute the MAE of both versions on all days and on public holidays.
  2. Run the DM–HLN test of the covariate version against the univariate one.
  3. Plot the daily MAE difference and mark the holidays.
  4. Interpretation: where does the gain come from?
- **Report**: four numbers, one DM statistic, one chart, one sentence
- Notebook: section B2

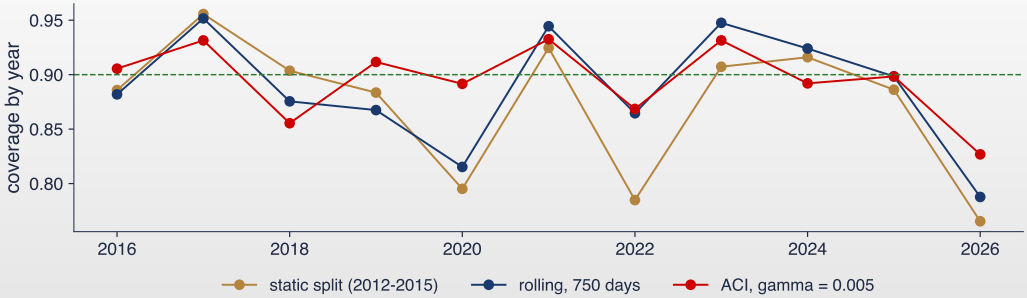
Back Exercise list

B3: conformal intervals for BET returns [Solved]

- **Question**: does a 90% interval calibrated on a quiet period keep its coverage when volatility changes?
- **Inputs**: BET daily log returns; forecast 0, score $|r_t|$; static split calibrated on 2012–2015, rolling (last 750 days), ACI with $\gamma = 0.005$; evaluation 2016–2026 ($n = 2675$)
- **Tasks**
  1. Compute the overall coverage of the three methods and the coverage by year.
  2. Run the Kupiec and Christoffersen tests on the miss sequences.
  3. Find the worst year of the static method.
  4. Interpretation: which property of returns breaks the static method?
- **Report**: three coverages, six p-values, one year, one sentence
- Notebook: section B3

Back Exercise list

B3: solution [Solved]



- Coverage: static 0.876, rolling 0.890, ACI 0.897; Kupiec $p < 0.001$, 0.081, 0.586; Christoffersen $p < 0.001$, < 0.001, < 0.001
- Static method by year: from 0.765 (2026) to 0.956; its threshold is fixed at 1.29%
- Interpretation: volatility clustering: the score $|r_t|$ is not exchangeable, so a quiet calibration under-covers in turbulent years; adaptive methods correct the frequency, but misses still cluster (Christoffersen)

ATS_ch13_seminar

B4: the ACI volatility design on the DAX [Proposed]

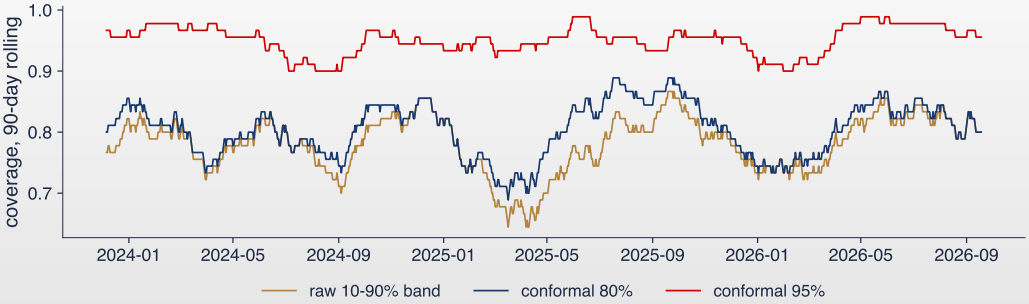
- **Question**: does the result of Gibbs and Candès (2021) hold for the DAX, and how does it depend on γ?
- **Inputs**: DAX daily returns; $V_t = r_t^2$, GARCH(1,1) on the previous 1250 days, score $|V_t - \hat{\sigma}_t^2|/\hat{\sigma}_t^2$, $\alpha = 0.1$; model: B3
- **Tasks**
  1. Compute the rolling (fixed-level) intervals and the ACI intervals for $\gamma = 0.001$, 0.005 and 0.02.
  2. Report the overall coverage, the range of the 500-day local coverage and the share of infinite intervals.
  3. Run the Christoffersen test for each method.
  4. Interpretation: which γ would you choose, and why?
- **Report**: one chart, four triples, four p-values, one sentence
- Notebook: section B4

B5: calibrating a foundation model on Bitcoin RV [Solved]

- **Question**: are the 10–90% bands of Chronos-Bolt for Bitcoin log realised variance calibrated, and can conformal prediction extend them to 95%?
- **Inputs**: Binance 5-minute RV, log scale, one day ahead from 2023, context 1024 days; online CQR (Prerequisites (14/23)) on the 10% and 90% quantiles, ACI on the last 250 scores, $\gamma = 0.005$
- **Tasks**
  1. Compute the coverage and the mean width of the raw band.
  2. Compute the coverage and the mean width of the conformal 80% and 95% intervals.
  3. Plot the 90-day rolling coverage of the three intervals.
  4. Interpretation: what does the conformal correction say about the model?
- **Report**: two numbers, four numbers, one chart, one sentence
- Notebook: section B5

Back Exercise list

B5: solution [Solved]



- Raw band: coverage 0.780, width 1.31; conformal 80%: 0.802, width 1.38; conformal 95%: 0.950, width 2.32 (log units, 1107 days from 8 September 2023)
- Interpretation: the deciles of a zero-shot model are close to calibrated on Bitcoin RV (78% for 80%); the conformal step adds the missing points and builds a 95% interval, 1.78 times the raw width, that the model cannot output

ATS_ch13_seminar

B6: EnbPI on the evening load peak [Proposed]

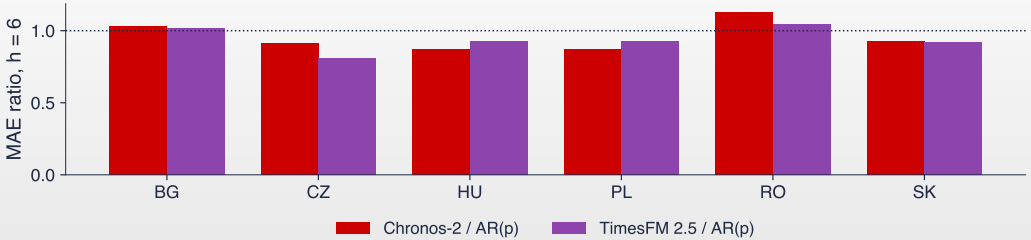
- **Question**: does a bootstrap ensemble with a sliding residual window give valid and sharp intervals for the 19:00 load?
- **Inputs**: Romanian load at 19:00, 2025–2026; EnbPI: ridge regressions on the expert ARX regressors, 20 block-bootstrap samples (blocks of 7 days) of the previous year; ACI around the expert ARX forecast (91-day window); model: B5
- **Tasks**
  1. Compute the 90% EnbPI intervals and the ACI intervals.
  2. Report coverage, mean width and the Christoffersen p-value of each.
  3. Plot both intervals and the load.
  4. Interpretation: which method would a grid operator prefer?
- **Report**: six numbers, one chart, one sentence
- Notebook: section B6

Back Exercise list

B7: CEE inflation, many countries, many tests [Solved]

- **Question**: do Chronos-2 and TimesFM beat AR for CEE inflation six months ahead, after correcting for multiple tests?
- **Inputs**: Eurostat HICP annual inflation, BG, CZ, HU, PL, RO, SK; monthly origins January 2012 – February 2026 (170 origins); $h = 6$; AR(p) by AIC; foundation models on the full history
- **Tasks**
  1. Compute the MAE ratio to AR by country and its geometric mean, for both models.
  2. Run the six country DM–HLN tests (HAC with 5 lags) and count rejections at 5%: raw, Holm, BH.
  3. Run the pooled DM test on the cross-country average loss differential.
  4. Interpretation: what is the honest one-line summary?
- **Report**: twelve ratios, two geometric means, six counts, two pooled tests, one sentence
- Notebook: section B7

B7: solution [Solved]



- Chronos-2 / AR: BG 1.03; CZ 0.91; HU 0.87; PL 0.87; RO 1.13; SK 0.92; geometric mean 0.952; rejections raw/Holm/BH: 1/0/0; pooled $t = -1.10$ ($p = 0.273$)
- TimesFM / AR: BG 1.02; CZ 0.81; HU 0.93; PL 0.93; RO 1.05; SK 0.92; geometric mean 0.938; rejections 0/0/0; pooled $t = -1.12$ ($p = 0.264$)
- Interpretation: both geometric means are below one, but no country difference survives Holm or BH and neither pooled test is significant: a modest average gain that these 14 years of six countries cannot confirm; Romania is where Chronos-2 does worst

ATS_ch13_seminar

B8: VaR of the WIG20 from foundation models [Proposed]

- **Question**: can a foundation model, alone or calibrated, deliver a VaR that passes the backtests of Chapter 9?
- **Inputs**: WIG20 daily returns, 2016–2026; GARCH(1,1)-t (1000 days, re-estimated every 250); Chronos-2 raw quantiles and with ACI; Chronos-Bolt 10% quantile shifted by a one-sided conformal threshold (ACI, window 500); model: B3 and B5
- **Tasks**
  1. Compute VaR 1% and VaR 5% with the four methods.
  2. Report hit rates, Kupiec and Christoffersen p-values.
  3. Compare the quantile losses with GARCH-t by a DM test.
  4. Interpretation: what does calibration fix, and what does it not fix?
- **Report**: a table of eight rows, one sentence
- Notebook: section B8

C1: a pre-registered inflation forecast [Proposed]

- **Question**: what will Romanian HICP inflation be in each of the next six months, with a calibrated 80% band, written down before the data exist?
- **Inputs**: Eurostat HICP, last observation August 2026 (6.3%); Chronos-2 medians and 10–90% bands from the last origin, widened horizon by horizon by the conformal threshold of the CQR scores of the last 60 origins; model: B5
- **Tasks**
  1. Write a five-line pre-registration: data vintage, model, horizons, scoring rule, what would count as a failure.
  2. Compute the six medians and the raw and conformal 80% bands.
  3. Add one covariate you believe matters and say how you would pass it to Chronos-2.
  4. Interpretation: how will you know in six months whether the forecast was good?
- **Report**: a five-line plan, a table of six months, a project plan
- Notebook: section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise foundation models and conformal prediction. The answer:
- (a) Chronos-Bolt gives a VaR 1% directly: request the 0.01 quantile.
- (b) Split conformal guarantees 90% coverage for every value of the covariate.
- (c) ACI guarantees that misses are independent over time.
- (d) With 19 calibration scores and alpha = 0.1, the threshold is the 18th smallest score.
- (e) Any stationary time series is exchangeable, so split conformal is exact for it.
- (f) The Holm correction controls the false discovery rate.
- (g) The geometric mean of relative errors ranks models the same way whatever baseline is used.
- (h) Tan et al. (2024) showed that time-series foundation models such as Chronos are not useful.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of eight verdicts with one line of justification each.

Key takeaways

- A conformal threshold is an order statistic: its coverage is exact on average and Beta-distributed given the calibration set
- Under dependence, feedback (ACI, quantile tracking) restores the long-run frequency for any sequence, not the independence of misses
- Foundation-model quantiles are scores to be calibrated; tokenisers and quantile heads limit what they can say
- Many series mean many tests: pool or correct before declaring a winner
- An AI answer is a draft: check quantile ranges, guarantees and the scope of cited studies

After the seminar

- Lecture 13 develops each topic of today
  - pretraining and model design; benchmarks and contamination
  - conformal theory and its time-series versions; calibration of foundation models
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: a live, pre-registered benchmark of foundation models on CEE inflation with conformal bands
- Reading: Angelopoulos and Bates (2023); Gibbs and Candès (2021); Romano, Patterson and Candès (2019); Ansari et al. (2024); Tan et al. (2024)
- **The seminar is for practice and is not graded**

References (1/2)

- Angelopoulos, A. N., & Bates, S. (2023). Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4), 494–591. doi:10.1561/2200000101
- Angelopoulos, A. N., Candès, E. J., & Tibshirani, R. J. (2023). Conformal PID control for time series prediction. *Advances in Neural Information Processing Systems 36*; *arXiv:2307.16895*. arXiv:2307.16895
- Ansari, A. F., Shchur, O., Küken, J., Auer, A., Han, B., Mercado, P., et al. (2025). Chronos-2: From univariate to universal forecasting. *arXiv:2510.15821*. arXiv:2510.15821
- Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., et al. (2024). Chronos: Learning the language of time series. *Transactions on Machine Learning Research*; *arXiv:2403.07815*. arXiv:2403.07815
- Barber, R. F., Candès, E. J., Ramdas, A., & Tibshirani, R. J. (2023). Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2), 816–845. doi:10.1214/23-AOS2276
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. doi:10.1016/0304-4076(86)90063-1
- Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862. doi:10.2307/2527341
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196. doi:10.1093/jjfinec/nbp001
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599

References (2/2)

- Fleming, P. J., & Wallace, J. J. (1986). How not to lie with statistics: The correct way to summarize benchmark results. *Communications of the ACM*, 29(3), 218–221. doi:10.1145/5666.5673
- Gibbs, I., & Candès, E. (2021). Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems 34*; *arXiv:2106.00170*. arXiv:2106.00170
- Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2), 281–291. doi:10.1016/S0169-2070(96)00719-4
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. jstor.org
- Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *The Journal of Derivatives*, 3(2), 73–84. doi:10.3905/jod.1995.407942
- Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R. J., & Wasserman, L. (2018). Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523), 1094–1111. doi:10.1080/01621459.2017.1307116
- Romano, Y., Patterson, E., & Candès, E. (2019). Conformalized quantile regression. *Advances in Neural Information Processing Systems 32*; *arXiv:1905.03222*. arXiv:1905.03222
- Tan, M., Merrill, M. A., Gupta, V., Althoff, T., & Hartvigsen, T. (2024). Are language models actually useful for time series forecasting? *Advances in Neural Information Processing Systems 37*; *arXiv:2406.16964*. arXiv:2406.16964
- Xu, C., & Xie, Y. (2021). Conformal prediction interval for dynamic time-series. *Proceedings of the 38th International Conference on Machine Learning (ICML)*, PMLR 139, 11559–11569. proceedings.mlr.press
- Ziel, F., & Weron, R. (2018). Day-ahead electricity price forecasting with high-dimensional structures: Univariate vs. multivariate modeling frameworks. *Energy Economics*, 70, 396–420. doi:10.1016/j.eneco.2017.12.016